

Vectors and geometry — Solutions

Warm-up

Question 1:

Let the position vectors of vertices A, B, C be $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively. Since D is the midpoint of AB and E is the midpoint of BC , we have:

$$\begin{aligned}\overrightarrow{AD} &= \frac{1}{2}\overrightarrow{AB} \\ \overrightarrow{BE} &= \frac{1}{2}\overrightarrow{BC}\end{aligned}$$

To find $\overrightarrow{DE}$, we use the vector path $D \rightarrow A \rightarrow C \rightarrow B \rightarrow E$ or simply $D \rightarrow B \rightarrow E$:

$$\begin{aligned}\overrightarrow{DE} &= \overrightarrow{DB} + \overrightarrow{BE} \\ &= \frac{1}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{BC} \\ &= \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{BC}) \\ &= \frac{1}{2}\overrightarrow{AC}\end{aligned}$$

(i) Since $\overrightarrow{DE} = \frac{1}{2}\overrightarrow{AC}$, the vector $\overrightarrow{DE}$ is a scalar multiple of $\overrightarrow{AC}$. This implies that DE is parallel to AC .

(ii) The length of DE is the magnitude of the vector $\overrightarrow{DE}$:

$$\begin{aligned}|\overrightarrow{DE}| &= \left|\frac{1}{2}\overrightarrow{AC}\right| \\ &= \frac{1}{2}|\overrightarrow{AC}|\end{aligned}$$

Thus, DE is half the length of AC .

Skill-building

Question 1:

Let the vertices of the parallelogram be A, B, C, D in order. Let $\overrightarrow{AB} = \mathbf{u}$ and $\overrightarrow{AD} = \mathbf{v}$. Then $\overrightarrow{DC} = \mathbf{u}$ and $\overrightarrow{BC} = \mathbf{v}$ because it is a parallelogram.

The diagonals are $\overrightarrow{AC}$ and $\overrightarrow{BD}$.

$$\begin{aligned}\overrightarrow{AC} &= \overrightarrow{AB} + \overrightarrow{BC} = \mathbf{u} + \mathbf{v} \\ \overrightarrow{BD} &= \overrightarrow{BA} + \overrightarrow{AD} = -\mathbf{u} + \mathbf{v}\end{aligned}$$

Now we calculate the squares of the lengths of the diagonals:

$$\begin{aligned}|\overrightarrow{AC}|^2 &= |\mathbf{u} + \mathbf{v}|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) \\ &= \mathbf{u} \cdot \mathbf{u} + 2(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v} \\ &= |\mathbf{u}|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2\end{aligned}$$

$$\begin{aligned}
|\overrightarrow{BD}|^2 &= |-\mathbf{u} + \mathbf{v}|^2 = (-\mathbf{u} + \mathbf{v}) \cdot (-\mathbf{u} + \mathbf{v}) \\
&= \mathbf{u} \cdot \mathbf{u} - 2(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v} \\
&= |\mathbf{u}|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2
\end{aligned}$$

Summing the squares of the diagonals:

$$\begin{aligned}
|\overrightarrow{AC}|^2 + |\overrightarrow{BD}|^2 &= (|\mathbf{u}|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2) + (|\mathbf{u}|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2) \\
&= 2|\mathbf{u}|^2 + 2|\mathbf{v}|^2
\end{aligned}$$

Since the sides of the parallelogram are $|\mathbf{u}|$, $|\mathbf{v}|$, $|\mathbf{u}|$, $|\mathbf{v}|$, the sum of the squares of the 4 sides is:

$$\text{Sum of squares of sides} = |\mathbf{u}|^2 + |\mathbf{v}|^2 + |\mathbf{u}|^2 + |\mathbf{v}|^2 = 2|\mathbf{u}|^2 + 2|\mathbf{v}|^2$$

Thus, the sum of the squares of the diagonals equals the sum of the squares of the four sides.

Easier Exam Questions

Question 1:

Given a rectangular prism with origin $O(0, 0, 0)$ and opposite vertex $P(a, b, c)$.

(i) The main diagonals connect opposite vertices. The vector $\mathbf{u}$ is the diagonal from O to P :

$$\mathbf{u} = \overrightarrow{OP} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Another main diagonal $\mathbf{v}$ connects the vertex $(a, 0, 0)$ to the vertex $(0, b, c)$:

$$\mathbf{v} = \begin{pmatrix} 0 - a \\ b - 0 \\ c - 0 \end{pmatrix} = \begin{pmatrix} -a \\ b \\ c \end{pmatrix}$$

(ii) $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if their dot product is zero:

$$\begin{aligned}
\mathbf{u} \cdot \mathbf{v} &= 0 \\
\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} -a \\ b \\ c \end{pmatrix} &= 0 \\
-a^2 + b^2 + c^2 &= 0 \\
b^2 + c^2 &= a^2
\end{aligned}$$

Wait, let's check the diagram's diagonal $\mathbf{v}$. If $\mathbf{v}$ is the diagonal from $(0, b, 0)$ to $(a, 0, c)$, then $\mathbf{v} = (a, -b, c)$. Then $\mathbf{u} \cdot \mathbf{v} = a^2 - b^2 + c^2 = 0 \implies b^2 = a^2 + c^2$. If $\mathbf{v}$ is from $(0, 0, c)$ to $(a, b, 0)$, then $\mathbf{v} = (a, b, -c)$. Then $\mathbf{u} \cdot \mathbf{v} = a^2 + b^2 - c^2 = 0 \implies a^2 + b^2 = c^2$.

Assuming the diagonal $\mathbf{v}$ is the one from $(0, 0, c)$ to $(a, b, 0)$ based on the prompt's requirement to show $a^2 + b^2 = c^2$:

$$\begin{aligned}
\mathbf{u} \cdot \mathbf{v} &= 0 \\
\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ -c \end{pmatrix} &= 0 \\
a^2 + b^2 - c^2 &= 0 \\
a^2 + b^2 &= c^2
\end{aligned}$$

(iii) Point Q is the intersection of the main diagonals, which is the midpoint of $\mathbf{u}$:

$$\mathbf{q} = \frac{1}{2}\mathbf{u} = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$$

For the diagonals to be perpendicular, we need $a^2 + b^2 = c^2$. A possible point Q would be any point where the coordinates of P satisfy this. For example, if $a = 3, b = 4, c = 5$, then $P(3, 4, 5)$ and $Q(1.5, 2, 2.5)$.

Question 2:

Given a tetrahedron ODEF with equilateral triangle faces of side length 1.

(i) Vertex O is $(0, 0, 0)$ and D is $(1, 0, 0)$. Since ODE is an equilateral triangle in the xy -plane, the x -coordinate of E is the midpoint of OD , and the y -coordinate is the height of the triangle:

$$\begin{aligned}
E_x &= \frac{1}{2} \\
E_y &= \sqrt{1^2 - (1/2)^2} = \sqrt{3/4} = \frac{\sqrt{3}}{2} \\
E_z &= 0
\end{aligned}$$

Thus, $E = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$.

(ii) Let $F = (x, y, z)$. Since F is the vertex of a regular tetrahedron, the projection of F onto the xy -plane is the centroid of $\triangle ODE$:

$$\text{Centroid } G = \frac{O + D + E}{3} = \left(\frac{0 + 1 + 1/2}{3}, \frac{0 + 0 + \sqrt{3}/2}{3}, 0\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}, 0\right)$$

Since F is above the xy -plane, $F = (1/2, \sqrt{3}/6, z)$. We use the fact that $|\overrightarrow{OF}| = 1$:

$$\begin{aligned}
\sqrt{(1/2)^2 + (\sqrt{3}/6)^2 + z^2} &= 1 \\
\frac{1}{4} + \frac{3}{36} + z^2 &= 1 \\
\frac{1}{4} + \frac{1}{12} + z^2 &= 1 \\
\frac{3+1}{12} + z^2 &= 1 \\
\frac{1}{3} + z^2 &= 1 \\
z^2 &= \frac{2}{3} \\
z &= \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}
\end{aligned}$$

Thus, $F = \left(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3}\right)$.

Question 3:

Given $OABC$ is a trapezium with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$, and $\overrightarrow{CB} = k\mathbf{a}$.

(i) First, find the position vectors of the midpoints M, N, P, Q :

$$\begin{aligned}\overrightarrow{OM} &= \frac{1}{2}\mathbf{a} \\ \overrightarrow{OQ} &= \frac{1}{2}\mathbf{c} \\ \overrightarrow{OB} &= \overrightarrow{OC} + \overrightarrow{CB} = \mathbf{c} + k\mathbf{a} \\ \overrightarrow{OP} &= \frac{\overrightarrow{OB} + \overrightarrow{OC}}{2} = \frac{(\mathbf{c} + k\mathbf{a}) + \mathbf{c}}{2} = \mathbf{c} + \frac{k}{2}\mathbf{a} \\ \overrightarrow{ON} &= \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2} = \frac{\mathbf{a} + (\mathbf{c} + k\mathbf{a})}{2} = \frac{1+k}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}\end{aligned}$$

Now, find the vectors $\overrightarrow{MP}$ and $\overrightarrow{QN}$:

$$\begin{aligned}\overrightarrow{MP} &= \overrightarrow{OP} - \overrightarrow{OM} = (\mathbf{c} + \frac{k}{2}\mathbf{a}) - \frac{1}{2}\mathbf{a} = \frac{k-1}{2}\mathbf{a} + \mathbf{c} \\ \overrightarrow{QN} &= \overrightarrow{ON} - \overrightarrow{OQ} = (\frac{1+k}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}) - \frac{1}{2}\mathbf{c} = \frac{1+k}{2}\mathbf{a}\end{aligned}$$

To show they bisect each other, we show the midpoint of MP is the same as the midpoint of QN :

$$\begin{aligned}\text{Midpoint of } MP &= \frac{\overrightarrow{OM} + \overrightarrow{OP}}{2} = \frac{\frac{1}{2}\mathbf{a} + (\mathbf{c} + \frac{k}{2}\mathbf{a})}{2} = \frac{1+k}{4}\mathbf{a} + \frac{1}{2}\mathbf{c} \\ \text{Midpoint of } QN &= \frac{\overrightarrow{OQ} + \overrightarrow{ON}}{2} = \frac{\frac{1}{2}\mathbf{c} + (\frac{1+k}{2}\mathbf{a} + \frac{1}{2}\mathbf{c})}{2} = \frac{1+k}{4}\mathbf{a} + \frac{1}{2}\mathbf{c}\end{aligned}$$

Since the midpoints are identical, MP and QN bisect each other.

(ii) If $MP \perp QN$, then $\overrightarrow{MP} \cdot \overrightarrow{QN} = 0$:

$$\begin{aligned}\left(\frac{k-1}{2}\mathbf{a} + \mathbf{c}\right) \cdot \left(\frac{1+k}{2}\mathbf{a}\right) &= 0 \\ \frac{(k-1)(k+1)}{4}|\mathbf{a}|^2 + \frac{1+k}{2}(\mathbf{a} \cdot \mathbf{c}) &= 0 \\ \frac{k^2-1}{4}|\mathbf{a}|^2 + \frac{1+k}{2}(\mathbf{a} \cdot \mathbf{c}) &= 0\end{aligned}$$

Divide by $\frac{1+k}{2}$ (since $k \neq -1$):

$$\begin{aligned}\frac{k-1}{2}|\mathbf{a}|^2 + \mathbf{a} \cdot \mathbf{c} &= 0 \\ \mathbf{a} \cdot \mathbf{c} &= \frac{1-k}{2}|\mathbf{a}|^2\end{aligned}$$

Now, consider $|\overrightarrow{OC}|^2$ and $|\overrightarrow{AB}|^2$:

$$\begin{aligned}
|\overrightarrow{OC}|^2 &= |\mathbf{c}|^2 \\
|\overrightarrow{AB}|^2 &= |\overrightarrow{OB} - \overrightarrow{OA}|^2 = |(\mathbf{c} + k\mathbf{a}) - \mathbf{a}|^2 = |\mathbf{c} + (k-1)\mathbf{a}|^2 \\
&= |\mathbf{c}|^2 + 2(k-1)(\mathbf{c} \cdot \mathbf{a}) + (k-1)^2|\mathbf{a}|^2
\end{aligned}$$

Substitute $\mathbf{a} \cdot \mathbf{c} = \frac{1-k}{2}|\mathbf{a}|^2$:

$$\begin{aligned}
|\overrightarrow{AB}|^2 &= |\mathbf{c}|^2 + 2(k-1) \left(\frac{1-k}{2}|\mathbf{a}|^2 \right) + (k-1)^2|\mathbf{a}|^2 \\
&= |\mathbf{c}|^2 - (k-1)^2|\mathbf{a}|^2 + (k-1)^2|\mathbf{a}|^2 \\
&= |\mathbf{c}|^2
\end{aligned}$$

Thus, $|\overrightarrow{OC}|^2 = |\overrightarrow{AB}|^2$.

Question 4:

Given $\mathbf{r}_1 = (1, 2, 1) + \lambda(1, 2, 2)$ and $\mathbf{r}_2 = (1, 2, 1) + \mu(2, 1, 2)$. Point A is $(1, 2, 1)$. Point D is $(5, 4, 5)$.

(i) Since D lies on $\mathbf{r}_2$, we check μ :

$$\begin{aligned}
(1, 2, 1) + \mu(2, 1, 2) &= (5, 4, 5) \\
2\mu &= 4 \implies \mu = 2
\end{aligned}$$

Thus, $\overrightarrow{AD} = \mu(2, 1, 2) = 2(2, 1, 2) = (4, 2, 4)$.

Since $ABCD$ is a rhombus, $|\overrightarrow{AB}| = |\overrightarrow{AD}|$. The vector $\overrightarrow{AB}$ must be a multiple of the direction vector of $\mathbf{r}_1$, which is $(1, 2, 2)$.

$$\overrightarrow{AB} = \lambda(1, 2, 2)$$

We find λ such that $|\overrightarrow{AB}| = |\overrightarrow{AD}| = \sqrt{4^2 + 2^2 + 4^2} = \sqrt{16 + 4 + 16} = 6$.

$$\begin{aligned}
|\lambda(1, 2, 2)| &= 6 \\
|\lambda|\sqrt{1^2 + 2^2 + 2^2} &= 6 \\
|\lambda|\sqrt{9} &= 6 \\
3|\lambda| &= 6 \\
|\lambda| &= 2 \implies \lambda = \pm 2
\end{aligned}$$

Possible coordinates of B :

$$\begin{aligned}
\text{If } \lambda &= 2, B = (1, 2, 1) + 2(1, 2, 2) = (3, 6, 5) \\
\text{If } \lambda &= -2, B = (1, 2, 1) - 2(1, 2, 2) = (-1, -2, -3)
\end{aligned}$$

(ii) The area of a rhombus is given by $|\overrightarrow{AB} \times \overrightarrow{AD}|$.

$$\begin{aligned}
\overrightarrow{AB} &= (2, 4, 4) \text{ (using } \lambda = 2) \\
\overrightarrow{AD} &= (4, 2, 4)
\end{aligned}$$

$$\begin{aligned}
\overrightarrow{AB} \times \overrightarrow{AD} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 4 & 4 \\ 4 & 2 & 4 \end{vmatrix} \\
&= \mathbf{i}(16 - 8) - \mathbf{j}(8 - 16) + \mathbf{k}(4 - 16) \\
&= (8, 8, -12)
\end{aligned}$$

$$\begin{aligned}
\text{Area} &= |(8, 8, -12)| = \sqrt{8^2 + 8^2 + (-12)^2} = \sqrt{64 + 64 + 144} = \sqrt{272} \\
&= \sqrt{16 \times 17} = 4\sqrt{17}
\end{aligned}$$

Harder Exam Questions

Question 1:

Let the position vectors of A, B, C be $\mathbf{a}, \mathbf{b}, \mathbf{c}$. In a rectangular prism, these are orthogonal vectors if they are the edges meeting at the origin O . Thus $\mathbf{a} \cdot \mathbf{b} = 0, \mathbf{a} \cdot \mathbf{c} = 0, \mathbf{b} \cdot \mathbf{c} = 0$.

Point P is the opposite vertex to O , so $\overrightarrow{OP} = \mathbf{a} + \mathbf{b} + \mathbf{c}$.

We examine $\triangle ACP$. The sides are $\overrightarrow{AC}, \overrightarrow{CP}, \overrightarrow{PA}$.

$$\begin{aligned}
\overrightarrow{AC} &= \mathbf{c} - \mathbf{a} \\
\overrightarrow{CP} &= \overrightarrow{OP} - \overrightarrow{OC} = (\mathbf{a} + \mathbf{b} + \mathbf{c}) - \mathbf{c} = \mathbf{a} + \mathbf{b} \\
\overrightarrow{PA} &= \overrightarrow{OA} - \overrightarrow{OP} = \mathbf{a} - (\mathbf{a} + \mathbf{b} + \mathbf{c}) = -\mathbf{b} - \mathbf{c}
\end{aligned}$$

To be a right-angled triangle, the dot product of two sides must be zero.

$$\text{Case 1: } \overrightarrow{AC} \cdot \overrightarrow{CP} = 0$$

$$\begin{aligned}
(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{a} + \mathbf{b}) &= 0 \\
\mathbf{c} \cdot \mathbf{a} + \mathbf{c} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} &= 0 \\
0 + 0 - |\mathbf{a}|^2 - 0 &= 0 \\
|\mathbf{a}|^2 &= 0
\end{aligned}$$

Since $\mathbf{a}$ is a non-zero vector, this is impossible.

$$\text{Case 2: } \overrightarrow{CP} \cdot \overrightarrow{PA} = 0$$

$$\begin{aligned}
(\mathbf{a} + \mathbf{b}) \cdot (-\mathbf{b} - \mathbf{c}) &= 0 \\
-\mathbf{a} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{c} - \mathbf{b} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{c} &= 0 \\
0 - 0 - |\mathbf{b}|^2 - 0 &= 0 \\
|\mathbf{b}|^2 &= 0
\end{aligned}$$

Since $\mathbf{b}$ is a non-zero vector, this is impossible.

$$\text{Case 3: } \overrightarrow{PA} \cdot \overrightarrow{AC} = 0$$

$$\begin{aligned}
(-\mathbf{b} - \mathbf{c}) \cdot (\mathbf{c} - \mathbf{a}) &= 0 \\
-\mathbf{b} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{a} - \mathbf{c} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} &= 0 \\
0 + 0 - |\mathbf{c}|^2 + 0 &= 0 \\
|\mathbf{c}|^2 &= 0
\end{aligned}$$

Since $\mathbf{c}$ is a non-zero vector, this is impossible. Thus, $\triangle ACP$ cannot be a right-angled triangle.

Question 2:

(i) $P(x, y, z)$ is on a sphere of radius 1, so $x^2 + y^2 + z^2 = 1$. We want to show $|x| + |y| + |z| \geq 1$.

$$\begin{aligned} (|x| + |y| + |z|)^2 &= x^2 + y^2 + z^2 + 2(|xy| + |yz| + |zx|) \\ &= 1 + 2(|xy| + |yz| + |zx|) \end{aligned}$$

Since $|xy|, |yz|, |zx| \geq 0$, it follows that:

$$\begin{aligned} (|x| + |y| + |z|)^2 &\geq 1 \\ |x| + |y| + |z| &\geq 1 \end{aligned}$$

(ii) This is the Cauchy-Schwarz inequality. We consider the dot product of $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$$

From the definition of the dot product, $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$.

$$|\mathbf{a} \cdot \mathbf{b}| = |\mathbf{a}||\mathbf{b}| |\cos \theta|$$

Since $|\cos \theta| \leq 1$, then $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}||\mathbf{b}|$

Substituting the components:

$$|a_1b_1 + a_2b_2 + a_3b_3| \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}$$

(iii) We want to show $|x| + |y| + |z| \leq \sqrt{3}$. Let $\mathbf{u} = (x, y, z)$ and $\mathbf{v} = (1, 1, 1)$.

$$\mathbf{u} \cdot \mathbf{v} = x(1) + y(1) + z(1) = x + y + z$$

Using the result from part (ii) with absolute values for x, y, z :

$$\mathbf{u} = (|x|, |y|, |z|) \text{ and } \mathbf{v} = (1, 1, 1)$$

$$\begin{aligned} ||x|(1) + |y|(1) + |z|(1)| &\leq \sqrt{|x|^2 + |y|^2 + |z|^2} \sqrt{1^2 + 1^2 + 1^2} \\ |x| + |y| + |z| &\leq \sqrt{x^2 + y^2 + z^2} \sqrt{3} \end{aligned}$$

Since $x^2 + y^2 + z^2 = 1$, then $|x| + |y| + |z| \leq 1 \cdot \sqrt{3} = \sqrt{3}$

Question 3:

Given $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$, and $\overrightarrow{PR} = k\overrightarrow{PQ}$.

(i) We find the position vectors of P and Q :

$$\overrightarrow{OP} = \frac{3}{5}\mathbf{a}$$

$$\overrightarrow{OQ} = 3\mathbf{b} \text{ (since } OB : OQ = 1 : 3)$$

Now, find $\overrightarrow{PQ}$:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = 3\mathbf{b} - \frac{3}{5}\mathbf{a}$$

Since $\overrightarrow{PR} = k\overrightarrow{PQ}$:

$$\begin{aligned}\overrightarrow{OR} &= \overrightarrow{OP} + \overrightarrow{PR} = \frac{3}{5}\mathbf{a} + k(3\mathbf{b} - \frac{3}{5}\mathbf{a}) \\ &= \frac{3}{5}\mathbf{a} - \frac{3k}{5}\mathbf{a} + 3k\mathbf{b} \\ &= \frac{3}{5}(1 - k)\mathbf{a} + 3k\mathbf{b}\end{aligned}$$

(ii) We are given $\overrightarrow{OR} = (1 - h)\mathbf{a} + h\mathbf{b}$. We equate the coefficients of $\mathbf{a}$ and $\mathbf{b}$:

$$\begin{aligned}\frac{3}{5}(1 - k) &= 1 - h \\ 3k &= h\end{aligned}$$

Substitute $h = 3k$ into the first equation:

$$\begin{aligned}\frac{3}{5}(1 - k) &= 1 - 3k \\ 3 - 3k &= 5 - 15k \\ 12k &= 2 \\ k &= \frac{2}{12} = \frac{1}{6}\end{aligned}$$

(iii) T lies on AQ and O, R, T are collinear. Thus $\overrightarrow{OT} = m\overrightarrow{OR}$ for some scalar m .

$$\begin{aligned}\overrightarrow{OT} &= m\left(\frac{3}{5}(1 - \frac{1}{6})\mathbf{a} + 3(\frac{1}{6})\mathbf{b}\right) \\ &= m\left(\frac{3}{5} \cdot \frac{5}{6}\mathbf{a} + \frac{1}{2}\mathbf{b}\right) \\ &= m\left(\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}\right)\end{aligned}$$

Since T lies on AQ , the sum of coefficients of $\mathbf{a}$ and $\mathbf{b}$ in $\overrightarrow{OT}$ must be 1:

$$\frac{1}{2}m + \frac{1}{2}m = 1 \implies m = 1$$

Thus, $\overrightarrow{OT} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$.

Question 4:

Given $A(4, 0, 0), C(0, 8, 0), T(0, 0, 6)$. The origin $O(0, 0, 0)$ is a vertex.

(i) The vertex B is the opposite vertex to O in the xy -plane, so $B(4, 8, 0)$. The diagonal BT connects $B(4, 8, 0)$ and $T(0, 0, 6)$:

$$\overrightarrow{BT} = \begin{pmatrix} 0 - 4 \\ 0 - 8 \\ 6 - 0 \end{pmatrix} = \begin{pmatrix} -4 \\ -8 \\ 6 \end{pmatrix}$$

Vector equation of line BT :

$$\mathbf{r} = \begin{pmatrix} 4 \\ 8 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -8 \\ 6 \end{pmatrix}$$

(ii) The center of the sphere is the midpoint of any main diagonal, such as BT :

$$\mathbf{c} = \frac{B + T}{2} = \left(\frac{4+0}{2}, \frac{8+0}{2}, \frac{0+6}{2} \right) = (2, 4, 3)$$

The radius r is half the length of BT :

$$|\overrightarrow{BT}| = \sqrt{(-4)^2 + (-8)^2 + 6^2} = \sqrt{16 + 64 + 36} = \sqrt{116} = 2\sqrt{29}$$

$$r = \frac{1}{2}|\overrightarrow{BT}| = \sqrt{29}$$

Sphere equation: $|\mathbf{v} - (2, 4, 3)| = \sqrt{29}$.

(iii) Line AM where $A(4, 0, 0)$ and $M(0, 4, 3)$:

$$\overrightarrow{AM} = \begin{pmatrix} 0-4 \\ 4-0 \\ 3-0 \end{pmatrix} = \begin{pmatrix} -4 \\ 4 \\ 3 \end{pmatrix}$$

$$\text{Line } AM : \mathbf{r}_{AM} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -4 \\ 4 \\ 3 \end{pmatrix}. \text{ Line } BT : \mathbf{r}_{BT} = \begin{pmatrix} 4 \\ 8 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -8 \\ 6 \end{pmatrix}.$$

To be skew, they must not be parallel and must not intersect. 1. Parallel check: Direction vectors $(-4, 4, 3)$ and $(-4, -8, 6)$ are not multiples of each other, so they are not parallel. 2. Intersection check: Equate components:

$$4 - 4s = 4 - 4\lambda \implies s = \lambda$$

$$0 + 4s = 8 - 8\lambda \implies 4s = 8 - 8s \implies 12s = 8 \implies s = 2/3$$

Check z -component:

$$0 + 3s = 0 + 6\lambda \implies 3(2/3) = 6(2/3) \implies 2 = 4 \text{ (False)}$$

Since they are not parallel and do not intersect, the lines AM and BT are skew.

Question 5:

Given a regular tetrahedron $OABC$ with side length 1.

(i) For any two vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ from the origin O , the angle between them is 60° because the faces are equilateral triangles.

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(60^\circ) = (1)(1)(1/2) = 1/2$$

Similarly, $\mathbf{b} \cdot \mathbf{c} = 1/2$ and $\mathbf{a} \cdot \mathbf{c} = 1/2$.

(ii) M divides AB in ratio $x : 1 - x$. Using the section formula:

$$\overrightarrow{OM} = (1 - x)\mathbf{a} + x\mathbf{b}$$

Now find the magnitude squared:

$$\begin{aligned} |\overrightarrow{OM}|^2 &= ((1 - x)\mathbf{a} + x\mathbf{b}) \cdot ((1 - x)\mathbf{a} + x\mathbf{b}) \\ &= (1 - x)^2|\mathbf{a}|^2 + 2x(1 - x)(\mathbf{a} \cdot \mathbf{b}) + x^2|\mathbf{b}|^2 \\ &= (1 - x)^2(1) + 2x(1 - x)(1/2) + x^2(1) \\ &= 1 - 2x + x^2 + x - x^2 + x^2 \\ &= 1 - x + x^2 \end{aligned}$$

Thus, $|\overrightarrow{OM}| = \sqrt{1 - x + x^2}$.

(iii) Let $f(x) = \frac{1+x-x^2}{1-x+x^2}$ for $0 < x < 1$. To prove $1 < f(x) \leq 5/3$:

$$f(x) - 1 = \frac{1 + x - x^2}{1 - x + x^2} - 1 = \frac{1 + x - x^2 - (1 - x + x^2)}{1 - x + x^2} = \frac{2x - 2x^2}{1 - x + x^2} = \frac{2x(1 - x)}{1 - x + x^2}$$

Since $0 < x < 1$, then $2x > 0$, $(1 - x) > 0$, and $1 - x + x^2$ (which has a negative discriminant) is always positive. Thus $f(x) - 1 > 0 \implies f(x) > 1$.

To prove $f(x) \leq 5/3$:

$$\begin{aligned} \frac{5}{3} - f(x) &= \frac{5}{3} - \frac{1 + x - x^2}{1 - x + x^2} \\ &= \frac{5(1 - x + x^2) - 3(1 + x - x^2)}{3(1 - x + x^2)} \\ &= \frac{5 - 5x + 5x^2 - 3 - 3x + 3x^2}{3(1 - x + x^2)} \\ &= \frac{2x^2 - 8x + 2}{3(1 - x + x^2)} \\ &= \frac{2(x^2 - 4x + 1)}{3(1 - x + x^2)} \end{aligned}$$

Wait, let's check the maximum of $f(x)$. $f'(x) = 0$ at $x = 1/2$. $f(1/2) = \frac{1+1/2-1/4}{1-1/2+1/4} = \frac{5/4}{3/4} = 5/3$.

Since the quadratic $x^2 - 4x + 1$ is not always positive, the maximum is indeed $5/3$ at $x = 1/2$.

(iv) $\angle MON$ is the angle between $\overrightarrow{OM}$ and $\overrightarrow{ON}$.

$$\cos(\angle MON) = \frac{\overrightarrow{OM} \cdot \overrightarrow{ON}}{|\overrightarrow{OM}| |\overrightarrow{ON}|}$$

Since M and N divide AB and BC symmetrically, $|\overrightarrow{OM}| = |\overrightarrow{ON}| = \sqrt{1 - x + x^2}$.

$$\begin{aligned} \overrightarrow{OM} \cdot \overrightarrow{ON} &= ((1 - x)\mathbf{a} + x\mathbf{b}) \cdot ((1 - x)\mathbf{b} + x\mathbf{c}) \\ &= (1 - x)^2(\mathbf{a} \cdot \mathbf{b}) + (1 - x)x(\mathbf{a} \cdot \mathbf{c}) + x(1 - x)(\mathbf{b} \cdot \mathbf{b}) + x^2(\mathbf{b} \cdot \mathbf{c}) \\ &= (1 - x)^2(1/2) + x(1 - x)(1/2) + x(1 - x)(1) + x^2(1/2) \\ &= \frac{1}{2}(1 - 2x + x^2 + x - x^2 + 2x - 2x^2 + x^2) \\ &= \frac{1}{2}(1 + x - x^2) \end{aligned}$$

$$\cos(\angle MON) = \frac{\frac{1}{2}(1+x-x^2)}{1-x+x^2} = \frac{1}{2}f(x)$$

Since $1 < f(x) \leq 5/3$, then $\frac{1}{2} < \cos(\angle MON) \leq \frac{5}{6}$.

$$\cos^{-1}(5/6) \leq \angle MON < \cos^{-1}(1/2) = \pi/3$$